

FILE

NAME: _____ TEACHER: _____

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS

PRELIMINARY HSC Year 11

ASSESSMENT TASK 2

July 2014

Time Allowed: 70 minutes

Total Marks: 56

General Instructions:

- Each question attempted is to be started on a NEW PAGE, clearly marked with the number of the question.
- Write using black or blue pen.
- APPROVED CALCULATORS may be used.
- All necessary working should be shown. Marks may be deducted if working is poorly set out or difficult to read.

Question 1:**(7 Marks)**

a) Factorise: $8x^3 - 64$ /2

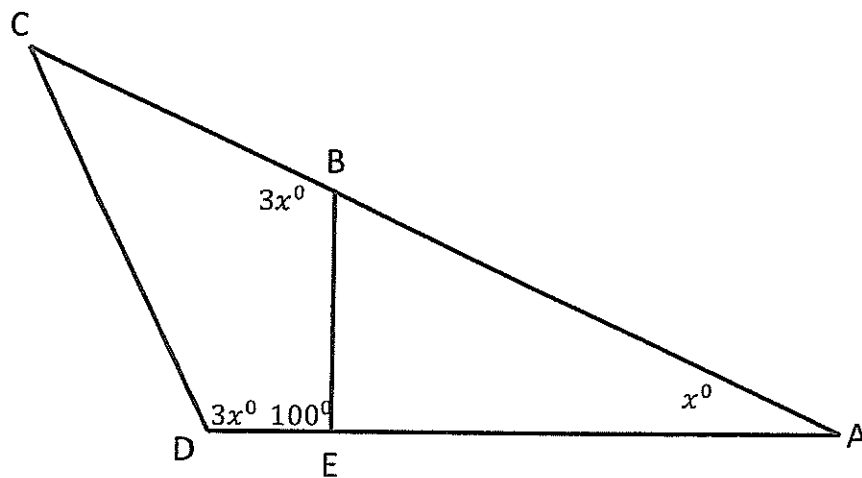
b) State the domain of: $f(x) = \frac{1}{\sqrt{9-x}}$ /2

c) Solve: $|x + 3| = 2x - 1$ /3

Question 2: Start a new page.**(7 Marks)**

- a) Find the equation of the line parallel to $2x - 3y + 5 = 0$, which passes through $(1, -2)$.
Give the answer in general form. /3

b)



- (i) Draw this diagram in your exam booklet

- (ii) Find the size of $\angle ACD$, Giving reasons for your statements. /2

c) Simplify: $\frac{4^3 \times 16^{(1-3n)}}{8^{-2n}}$ /2

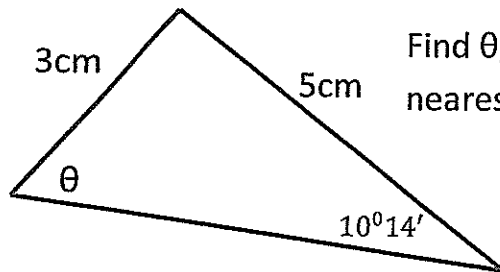
Question 3: Start a new page.

(7 Marks)

a) Find the exact value of: $\sin 225^\circ$

/2

b)



Find θ , to the nearest minute.

Diagram not to Scale

/3

c) Solve: $\cos \theta = \frac{-\sqrt{3}}{2}$, $0^\circ \leq \theta \leq 360^\circ$

/2

Question 4: Start a new page.

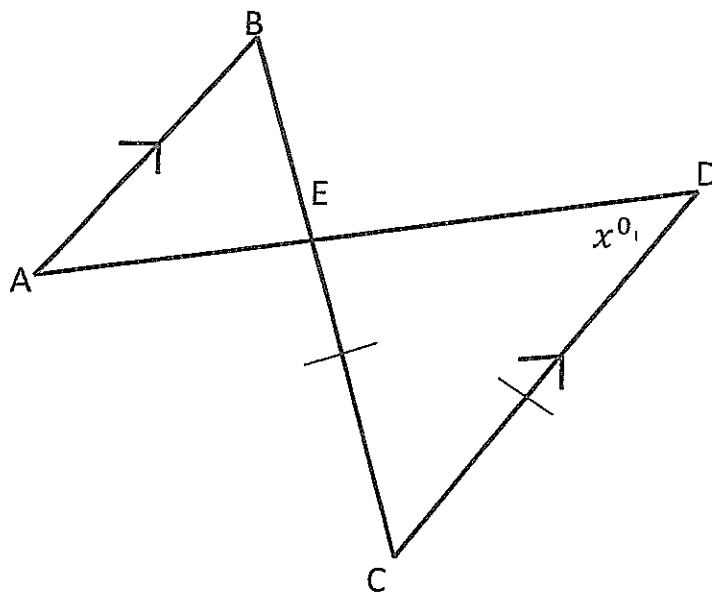
(7 Marks)

a) If $\tan A = \frac{\sqrt{2}}{3}$, and $\cos A < 0$

Find the exact value of $\sin A$

/2

b)



(i) Copy this diagram into your exam booklet

/3

(ii) Prove: $AB = BE$

c) Prove that: $\tan^2 \theta \operatorname{cosec}^2 \theta = 1 + \tan^2 \theta$

/2

Question 5 Start a new page

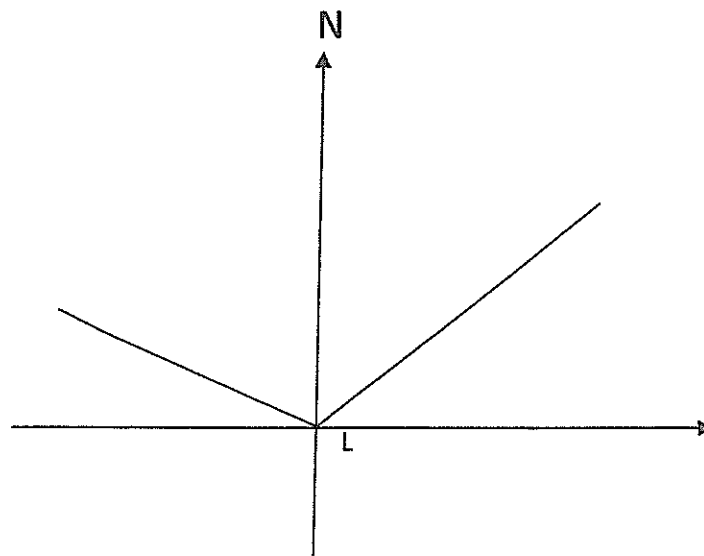
(7 Marks)

a) Simplify: $\frac{\sin(90-\theta)}{\sin \theta}$ /1

b) From a lighthouse, L, a ship, S, bears 053° T, and is at a distance of 8 nautical miles.

From L, a boat, B, bears 293° T and is at a distance of 6 nautical miles.

- (i) Copy the diagram into your answer booklet and show all the given information.

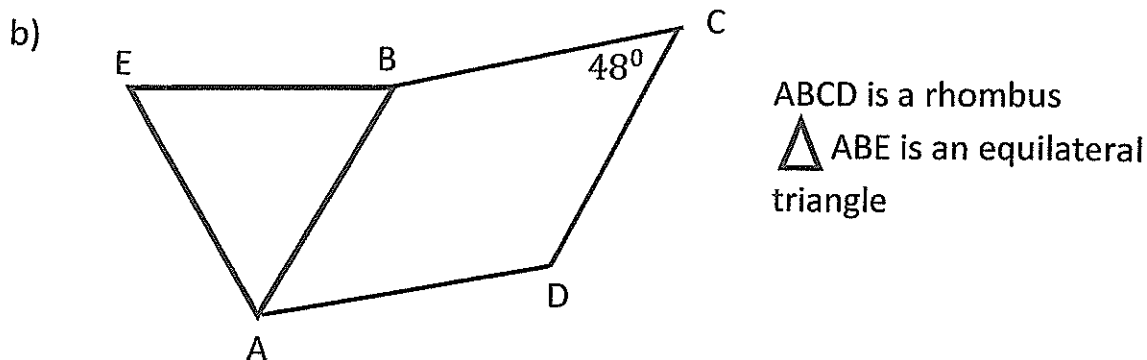


- (ii) Show that $\angle SLB$ is 120°
- (iii) Find the exact distance of the ship from the boat, in simplest surd form.
- (iv) Find the bearing of the ship from the boat to the nearest degree. /6

Question 6: Start a new page.

(7 Marks)

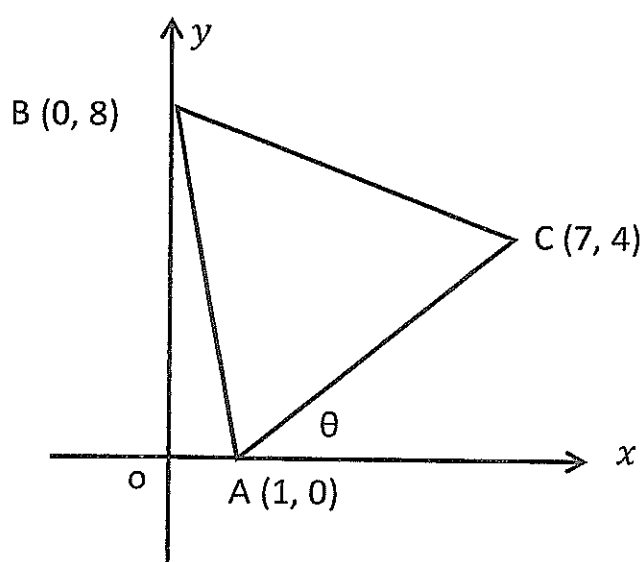
a) Solve $4\sin^2\theta - 3 = 0$, $0^\circ \leq \theta \leq 360^\circ$ /3



- (i) Draw this diagram in your exam booklet
- (ii) Find the size of $\angle EAD$, giving reasons for your statement.
- (iii) Find the size of $\angle EDA$ giving reasons for your statement. /4

Question 7: Start a new page

(7 Marks)



- (i) Find the gradient of AC /7
- (ii) Find θ to the nearest degree.
- (iii) Find the co-ordinates of D, the midpoint of AC
- (iv) Show that AC is perpendicular to BD

(v) What does (iv) say about $\triangle ABC$.

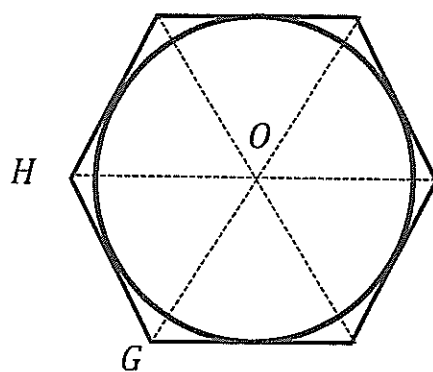
(vi) Write the co-ordinates of the point E, such that ABCE is a rhombus.

Question 8: Start a new page

(7 Marks)

a) One of the exterior angles in a regular polygon is 15° . /1
How many sides does it have?

b)



A regular hexagon is drawn outside a circle of radius, r , centre, O

/6

(i) Show that $\triangle OGH$ is equilateral

(ii) Show that the area of $\triangle OGH$ is given by

$$A = \frac{\sqrt{3}r^2}{3}$$

(iii) Find the area of the hexagon in terms of r .

(iv) By considering the result in (iii)

Show that $\pi < 2\sqrt{3}$

End of Test

Question 1

$$\begin{aligned} \text{a) } 8x^3 - 64 &= 8(x^3 - 8) \\ &= 8(x-2)(x^2 + 2x + 4) \end{aligned}$$

$$\text{b) } x < 9$$

$$\text{c) } |x+3| = 2x-1$$

$$x+3 = 2x-1$$

$$-x = -4$$

$$x = 4$$

$$-(x+3) = 2x-1$$

$$-x-3 = 2x-1$$

$$-3x = 2$$

$$x = -\frac{2}{3}$$

Test

$$x = 4 \checkmark$$

$$x = -\frac{2}{3} \times$$

$$\therefore x = 4$$

Question 2

$$\text{a) } 2x - 3y + 5 = 0$$

$$3y = 2x + 5$$

$$y = \frac{2}{3}x + \frac{5}{3}$$

m of any line parallel is $\frac{2}{3}$

$$\therefore y + 2 = \frac{2}{3}(x - 1)$$

$$3y + 6 = 2x - 2$$

$$2x - 3y - 8 = 0$$

$$\begin{aligned} \text{c) } \frac{4^3 \times 16^{(1-3n)}}{8^{-2n}} &= \frac{2^4 \times 2^{4-12n}}{2^{-6n}} \\ &= \frac{2^{10-12n}}{2^{-6n}} \\ &= 2^{10-6n} \end{aligned}$$

$$\text{b) } \angle ACD = 180 - 4x \text{ (angle sum } \triangle ACD)$$

$$\angle ACD = 360 - 100 - 6x \text{ (angle sum quad)}$$

$$180 - 4x = 260 - 6x$$

$$2x = 80$$

$$x = 40^\circ$$

$$\therefore \angle ACD = 180^\circ - 160^\circ = 20^\circ$$

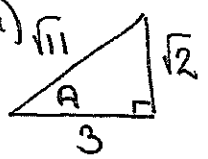
Question 3

a) $\sin 225^\circ = \sin (180^\circ + 45^\circ)$
 $= -\sin 45^\circ$
 $= -\frac{1}{\sqrt{2}}$ 2

b) $\frac{\sin \theta}{5} = \frac{\sin 10^\circ 14'}{3}$
 $\sin \theta = \frac{5 \sin 10^\circ 14'}{3}$
 $\theta = 17^\circ 13', 162^\circ 47'$ 3

c) θ lies in quad 2 or 3
 $\therefore \theta = 180^\circ - 30^\circ, 180^\circ + 30^\circ$
 $= 150^\circ, 210^\circ$ 2

Question 4

a)  A lies in quad 3
 $\therefore \sin A = -\frac{\sqrt{2}}{\sqrt{11}}$ 2

b) $\angle CED = x^\circ$
(angles opp. equal sides)
 $\therefore \angle BEA = x^\circ$ (vertically opp angles)
 $\angle BAE = x^\circ$ (alternate angles AB || DC)
 $\therefore AB = BE$ (sides opp equal angles) 3

c) $\tan^2 \theta \operatorname{cosec}^2 \theta = 1 + \tan^2 \theta$

LHS

$$\frac{\sin^2 \theta}{\cos^2 \theta} \times \frac{1}{\sin^2 \theta} = \frac{1}{\cos^2 \theta}$$

$$= \sec^2 \theta$$

$$= 1 + \tan^2 \theta$$

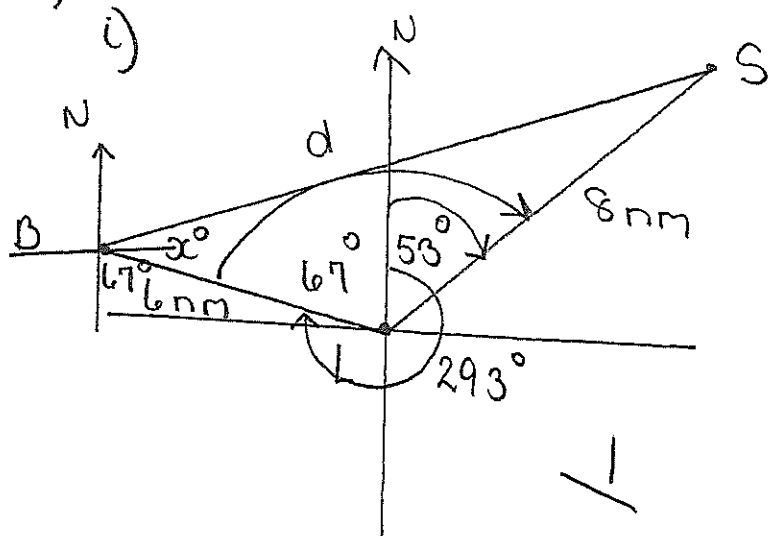
$$= \text{RHS}$$
 2

Question 5

a) $\cot \theta$

b)

i)



$$\text{ii) } 67^\circ + 53^\circ = 120^\circ$$

$$\text{iii) } d^2 = 6^2 + 8^2 - 2 \times 6 \times 8 \times \cos 120^\circ$$

$$d^2 = 148$$

$$d = \sqrt{148} = 2\sqrt{37} \text{ nm}$$

$$\text{iv) } \frac{\sin x}{8} = \frac{\sin 120^\circ}{2\sqrt{37}}$$

$$x^\circ = 34^\circ 42'$$

$\therefore$ bearing of ship from boat

$$= 180^\circ - 67^\circ - 35^\circ$$

$$= 078^\circ \text{ T}$$

Question 6

a) $4 \sin^2 \theta = 3$

$$\sin \theta = \pm \frac{\sqrt{3}}{2}$$

$$\theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ$$

b) ii) $\angle BAD = 48^\circ$ (opp angles of rhombus)
 $\therefore \angle EAD = 48^\circ + 60^\circ \rightarrow$ (equilateral triangle)
 $= 108^\circ$

iii) $EA = AD$ rhombus and equilateral triangle
 $\therefore \triangle EAD$ is isosceles

$\therefore \angle EDA = \frac{180^\circ - 108^\circ}{2}$ (angles opposite equal sides)
 $= 36^\circ$

Question 7

i) $m_{AC} = \frac{4}{6}$
 $= \frac{2}{3}$

ii) $\tan \theta = \frac{2}{3}$
 $\theta = 34^\circ$

iii) $D = (4, 2)$

iv) $m_{BD} = \frac{8-2}{0-4}$
 $= \frac{6}{-4}$
 $= -\frac{3}{2}$

v) isosceles

vi) $(8, -4)$

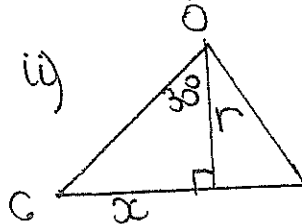
as $m_{BD} \times m_{AC} = -1$

$AC \perp BD$

Question 8

a) $360^\circ \div 15^\circ$
 $= 24 \text{ sides}$

b) i) $\angle COH = 60^\circ$ regular hexagon
 $OH = OG$ regular hexagon
 $\therefore \angle OHC = \angle OGH = 60^\circ$ (angles opp equal sides)
 $\therefore$ equilateral



$\tan 30^\circ = \frac{1}{\sqrt{3}}$

$\frac{1}{\sqrt{3}} = \frac{x}{r}$

$x = \frac{r}{\sqrt{3}}$

$\therefore \text{area} = \frac{1}{2} \times \frac{2r}{\sqrt{3}} \times r = \frac{\sqrt{3}r^2}{3}$

iii) area of hexagon
 $= 6 \times \frac{\sqrt{3}r^2}{3}$
 $= 2\sqrt{3}r^2$ ✓

iv) area of circle < area of hexagon
 $\pi r^2 < 2\sqrt{3}r^2$
 $\pi < 2\sqrt{3}$ ✓
